



INQUISITORS AI & QUIZ SOCIETY

UET LAHORE

Web Development Internship

Week 1 | Complete HTML Notes

A full, easy-to-revise guide covering every core HTML concept — document structure, tags & attributes, text formatting, links & images, tables, forms, semantic HTML, media, accessibility, and best practices — explained with tables, examples, key points, interview questions, practice tasks, and a visual diagram on every page.

CHAIRMAN

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Keep Learning • Keep Growing

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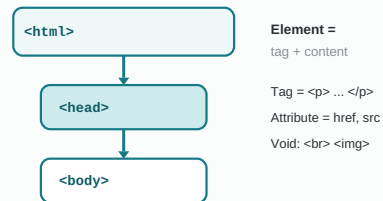
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1. Introduction to HTML

HTML (HyperText Markup Language) is the standard language used to create the structure of every webpage. Browsers read HTML and render it as a visual page.

Term	Meaning
HTML	HyperText Markup Language — structures web content
Element	A tag plus its content, e.g. <code><p>Hello</p></code>
Tag	The markup itself, e.g. <code><p></code> (opening) and <code></p></code> (closing)
Attribute	Extra info added inside a tag, e.g. href, src, class
Browser	Software that reads HTML/CSS/JS and displays the page
Void Element	Tag with no closing tag, e.g. <code>
</code> , <code></code> , <code><input></code>

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Element =
tag + content

Tag = `<p> ... </p>`

Attribute = href, src

Void: `<br>` `<img>`

Created by Tim Berners-Lee, 1991

HTML = HyperText Markup Language

Structure of every webpage

Example

```
<p>This is an element</p>
<br>                                <!-- void element, no closing tag -->
 <!-- void element with an attribute -->
```

Key Points

- HTML is not a programming language — it has no logic, only structure
- HTML works together with CSS (styling) and JavaScript (behaviour)
- Most tags come in pairs: opening & closing

Interview Question

Q: What is the difference between an element and a tag?

Ans: A tag is the markup symbol itself; an element is the tag together with its content and closing tag.

Practice Task

List 5 HTML tags you already know and identify which are void elements.

2. HTML Document Structure

Every HTML page follows a fixed skeleton that tells the browser how to read and render the page.

Tag	Description	Use
<!DOCTYPE html>	Declares document as HTML5	Always the first line of the file
<html>	Root element wrapping the whole page	Contains <head> and <body>
<head>	Holds meta info, not shown on page	Title, meta tags, links to CSS/JS
<title>	Sets the browser tab title	SEO and identification
<body>	Holds all visible content	Everything the user sees
<meta charset>	Sets character encoding	Correct display of text/symbols

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Example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Page</title>
</head>
<body>
  <h1>Hello World</h1>
</body>
</html>
```

Key Points

- Doctype must be the first line
- head = metadata, body = visible content
- Viewport meta tag is essential for responsiveness

Interview Question

Q: Why do we write <!DOCTYPE html>?

Ans: To tell the browser to render the page using HTML5 standards mode.

Practice Task

Create a blank HTML5 boilerplate file with proper meta tags and a title of your choice.

3. Elements, Tags & Global Attributes

Elements can be block-level or inline, and almost every tag accepts a set of global attributes that work everywhere.

Concept	Description	Example
Block-level	Starts on a new line, takes full width	<div>, <p>, <h1>, <table>
Inline	Sits within the text flow	, <a>, ,
id	Unique identifier for one element	id="header"
class	Reusable identifier for styling groups	class="card"
style	Inline CSS on a single element	style="color:red"
title / lang	Tooltip text / content language	title="..." lang="en"

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<div> block-level (full width)

This is a paragraph with an inline highlighted word inside.

id="intro"

class="card"

id → unique, one element
class → reusable, many elements
style="color:teal" → inline CSS

Example

```
<p id="intro" class="highlight" title="Intro paragraph">
  Welcome to <span style="color:teal;">HTML</span>!
</p>
```

Key Points

- Global attributes can be used on almost any HTML tag
- id must be unique on the page, class can repeat
- Block elements stack vertically, inline elements sit side by side

Interview Question

Q: Can two elements share the same id?

Ans: No. An id must be unique within the page; use class when multiple elements need the same identifier.

Practice Task

Create three paragraphs: give one an id, give two of them the same class, and style that class differently.

4. Headings & Paragraphs

Headings define the hierarchy of a page's content, and paragraphs hold blocks of readable text.

Tag	Description	Use
<h1>	Most important heading	Main page title (ideally one per page)
<h2> - <h6>	Decreasing importance	Sub-headings, section titles
<p>	Paragraph of text	Body content
<hr>	Horizontal rule / divider line	Separates sections
 	Line break	Forces a new line inside text

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H1 Web Development

H2 Introduction

H3 Sub Topic

p Body paragraph text goes here...

<hr> horizontal divider

line one

line two

Order:
h1 → h2 → h3...
(never skip levels)

Example

```
<h1>Web Development</h1>
<h2>Introduction</h2>
<p>HTML is the foundation of every website.<br>
It defines structure and content.</p>
<hr>
<h2>Next Topic</h2>
```

Key Points

- Headings should be used in logical order (h1 → h2 → h3...), not for making text bigger
- br breaks a line without starting a new paragraph
- hr is a semantic divider, not just a styling line

Interview Question

Q: Why shouldn't headings be used only to make text look bigger?

Ans: Headings define document structure/outline (important for SEO & accessibility); font size should be controlled with CSS instead.

Practice Task

Create a page with one h1, two h2 sub-sections, and a short paragraph under each.

5. Text Formatting Tags

HTML provides inline tags to change the meaning or appearance of specific pieces of text.

Tag	Description	Use
<code> / </code>	Bold: style-only / semantic importance	Emphasis vs. real importance
<code><i> / </code>	Italic: style-only / semantic emphasis	Alternate voice vs. emphasised meaning
<code><u> / <mark></code>	Underlined / highlighted text	Annotation, search results
<code><small></code>	Smaller text	Fine print, disclaimers
<code><sub> / <sup></code>	Subscript / superscript	H ₂ O, x ²
<code> / <ins></code>	Deleted / inserted text	Show edits, strikethrough price

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Warning: This is very important.

Formula: H₂O and E=mc²

Price: ~~\$50~~ \$35

Deadline: Friday

`<strong>/<em>` = meaning
`<b>/<i>` = style only
`<mark>` = highlight
`<del>/<ins>` = edits

Example

```
<p><strong>Warning:</strong> This is <em>very</em> important.</p>
<p>Formula: H<sub>2</sub>O and E=mc<sup>2</sup></p>
<p>Price: <del>$50</del> <ins>$35</ins></p>
<p><mark>Deadline: Friday</mark></p>
```

Key Points

- strong/em carry meaning, b/i are style-only
- sub/sup are used for chemical formulas and exponents
- del/ins are great for showing price changes or tracked edits

Interview Question

Q: What is the difference between `<b>` and `<strong>`?

Ans: Both render as bold, but strong signals real importance to browsers/screen readers, while b is purely visual.

Practice Task

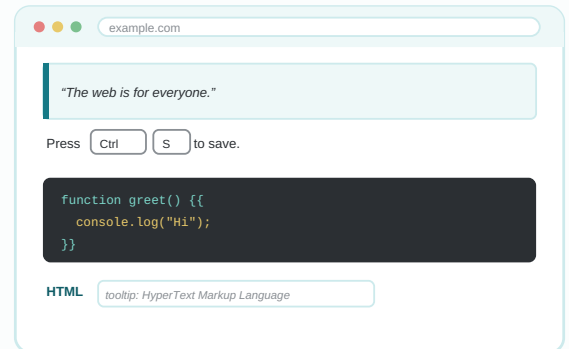
Write a paragraph using strong, em, mark, and sub/sup at least once each.

6. Quotations, Code & Preformatted Text

These tags mark up quoted content, inline code, code blocks, and text where whitespace matters.

Tag	Description	Use
<blockquote>	Long, block-level quotation	Quoting another source
<q>	Short, inline quotation	Adds quotation marks automatically
<cite>	Title of a work being referenced	Book/article/author name
<code> / <pre>	Inline code / preformatted block	Variable names / preserves spacing
<kbd>	Keyboard input	Shows a key the user should press
<abbr>	Abbreviation with tooltip meaning	HTML, CSS, etc. with title attribute

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Example

```
<blockquote cite="https://example.com">  
  The web is for everyone.  
</blockquote>  
<p>Press <kbd>Ctrl</kbd> + <kbd>S</kbd> to save.</p>  
<pre>  
function greet() {  
  console.log("Hi");  
}  
</pre>  
<p><abbr title="HyperText Markup Language">HTML</abbr> is fun.</p>
```

Key Points

- pre keeps every space and line break exactly as typed
- blockquote is block-level, q is inline
- abbr + title gives a hover tooltip and helps accessibility

Interview Question

Q: When would you use pre instead of a normal paragraph?

Ans: When whitespace and line breaks must be preserved exactly, such as displaying source code.

Practice Task

Add a blockquote with a citation, a line of code using code/pre, and one abbreviation with a tooltip.

7. Lists

Lists organise related items using bullets, numbers, or term/definition pairs.

Tag	Description	Use
	Unordered (bulleted) list	Order doesn't matter
	Ordered (numbered) list	Order matters, e.g. steps
	A single list item	Used inside ul or ol
<dl>/<dt>/<dd>	Description list / term / details	Glossaries, FAQs
Nested list	A list inside a list item	Sub-categories, multi-level menus

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1. Plan the layout

2. Write the HTML

- Head section
- Body section

3. Add CSS styling

HTML

Structures web content

CSS

Styles the HTML content

<dl><dt><dd>

Example

```
<ol>
  <li>Plan the layout</li>
  <li>Write the HTML
    <ul>
      <li>Head section</li>
      <li>Body section</li>
    </ul>
  </li>
  <li>Add CSS styling</li>
</ol>

<dl>
  <dt>HTML</dt>
  <dd>Structures web content</dd>
</dl>
```

Key Points

- ul = no order, ol = order matters
- Lists can be nested inside a single
- dl/dt/dd are perfect for glossaries or FAQs

Interview Question

Q: When should you use ol instead of ul?

Ans: When the sequence of items matters, such as step-by-step instructions or rankings.

Practice Task

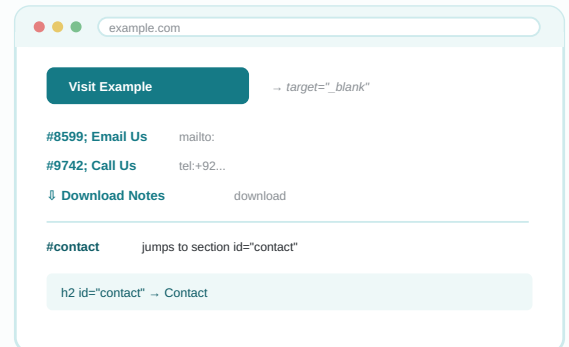
Create a nested list: a numbered list of 3 steps, where step 2 contains a bulleted sub-list of 2 items.

8. Links (Anchor Tag)

The anchor tag creates hyperlinks to other pages, sections, files, emails, or phone numbers.

Attribute/Type	Description	Use
href	Destination of the link	URL, file path, or #id
target="_blank"	Opens link in a new tab	External links
Internal link	Links to another section on the same page	href="#section-id"
mailto: / tel:	Opens email client / phone dialer	href="mailto:..." / href="tel:..."
download	Forces the file to download	href="file.pdf" download

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Example

```
<a href="https://example.com" target="_blank">Visit Site</a>
<a href="#contact">Jump to Contact Section</a>
<a href="mailto:info@inquisitors.com">Email Us</a>
<a href="tel:+923096888664">Call Us</a>
<a href="notes.pdf" download>Download Notes</a>

<h2 id="contact">Contact</h2>
```

Key Points

- href is required; without it, the tag is not really a link
- Use target="_blank" for external sites so users don't lose your page
- #id links jump to any element that has a matching id

Interview Question

Q: How do you make a link open in a new browser tab?

Ans: Add the attribute target="_blank" to the anchor tag.

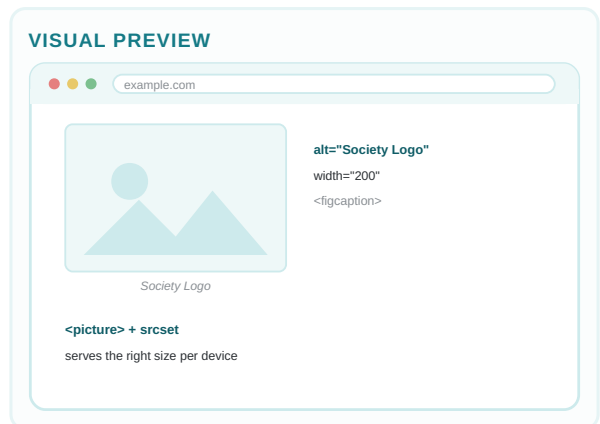
Practice Task

Create a navigation menu with 3 anchor links: one internal (#id jump), one external (new tab), and one mailto link.

9. Images

Images are embedded with the `img` tag, and can be made responsive or given captions using extra tags.

Tag/Attribute	Description	Use
<code></code>	Embeds an image	<code>src</code> = file path, <code>alt</code> = fallback text
<code>width / height</code>	Sets displayed size	Prevents layout shift while loading
<code><figure>/<figcaption></code>	Groups an image with a caption	Semantic, described image block
<code><picture> / srcset</code>	Multiple image sources / resolutions	Responsive, sharper images



Example

```
<figure>
  
  <figcaption>Inquisitors AI & Quiz Society Logo</figcaption>
</figure>

<picture>
  <source media="(max-width:600px)" srcset="small.jpg">
  
</picture>
```

Key Points

- `alt` text is mandatory for accessibility and SEO
- `figure` + `figcaption` is the semantic way to caption an image
- `picture/srcset` serve the right image size per device

Interview Question

Q: Why should every `img` tag include an `alt` attribute?

Ans: It provides a text description for screen readers and displays if the image fails to load, and it helps SEO.

Practice Task

Add an image inside a `figure` tag with a proper `figcaption` describing it.

10. Tables — Basics

Tables display data in a grid of rows and columns using a structured set of tags.

Tag	Description	Use
<table>	Wraps the whole table	Container for rows/columns
<tr>	Table row	One row of data
<th>	Header cell (bold, centered)	Column/row labels
<td>	Standard data cell	Regular values
border (CSS)	Adds visible grid lines	Readability, not an HTML attribute anymore

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Name	Age	Country
John	25	USA
Sara	28	UK

<table><tr><th><td>

Example

```
<table>
  <tr><th>Name</th><th>Score</th></tr>
  <tr><td>Ali</td><td>90</td></tr>
  <tr><td>Sara</td><td>95</td></tr>
</table>
```

Key Points

- th is for headers, td is for data
- Every tr should have the same number of cells
- Table styling (borders, spacing) is best done with CSS

Interview Question

Q: What is the difference between th and td?

Ans: th defines a header cell (bold & centered by default), td defines a normal data cell.

Practice Task

Create a simple table with 2 columns and 3 rows of your own data.

11. Tables — Advanced

Larger tables use extra structural tags to group rows and merge cells for complex layouts.

Tag/Attribute	Description	Use
<thead>/<tbody>/<tfoot>	Groups header/body/footer rows	Improves structure & styling
<caption>	Table title	Placed right after <table>
colspan	Merges cells across columns	Wide header spanning columns
rowspan	Merges cells across rows	One value spanning multiple rows

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example.com

Name	Days (colspan=2)	
Ali	Mon	Tue
End of Report (colspan=3)		

thead / tbody / tfoot

colspan merges columns

rowspan merges rows

Example

```
<table>
  <caption>Weekly Attendance</caption>
  <thead><tr><th>Name</th><th colspan="2">Days</th></tr></thead>
  <tbody><tr><td>Ali</td><td>Mon</td><td>Tue</td></tr></tbody>
  <tfoot><tr><td colspan="3">End of Report</td></tr></tfoot>
</table>
```

Key Points

- thead/tbody/tfoot group rows logically and help CSS/JS target them
- colspan merges columns, rowspan merges rows
- caption gives the table an accessible title

Interview Question

Q: When would you use rowspan?

Ans: When one value in a row needs to visually span and apply to multiple rows below it, e.g. a shared category.

Practice Task

Create a table with a caption, a header row using colspan, and at least 3 data rows.

12. Forms — Structure & Basic Inputs

Forms collect user input. Every form starts with the form tag and a chosen submission method.

Tag/Attribute	Description	Use
<form>	Wraps all input controls	action + method define submit behaviour
action	URL the form data is sent to	Server endpoint
method="get" / "post"	Sends data via URL / request body	Visible search vs. hidden sensitive data
input type="text/ email/password"	Single-line, validated, or masked input	Name, email address, passwords

Sign Up Form

method="post" action="/submit"

Username

JohnDoe123

Email

john@example.com

Password

Submit

Reset

Example

```
<form action="/submit" method="post">
  <input type="text" name="username" placeholder="Username">
  <input type="email" name="email" placeholder="Email">
  <input type="password" name="pass" placeholder="Password">
</form>
```

Key Points

- GET shows data in the URL, POST hides it in the request body
- Always give inputs a name attribute so their value gets submitted
- placeholder is just a hint, not a real value

Interview Question

Q: Difference between GET and POST in a form?

Ans: GET sends data via the URL (visible, limited size), POST sends it in the request body (hidden, larger data).

Practice Task

Build a simple sign-up form with username, email, and password fields using the post method.

13. Forms — HTML5 Input Types

HTML5 introduced many specialised input types that give built-in validation and better mobile keyboards.

Input Type	Use	Example
tel / date / time	Phone / date / time selection	<code><input type="date"></code>
number / range	Numeric input / slider	<code><input type="number" min="1" max="10"></code>
color / search / file	Colour picker / search field / upload	<code><input type="color"></code>
checkbox / radio	Multiple selections / single selection	<code>name="gender" groups radios</code>

Example

```
<input type="date">
<input type="range" min="0" max="100">
<input type="color">
<input type="checkbox" name="agree"> I agree
<input type="radio" name="plan" value="basic"> Basic
<input type="radio" name="plan" value="pro"> Pro
```

Key Points

- Radio buttons need the same name attribute to work as a group
- Checkboxes allow multiple selections; radios allow only one
- Specialised input types provide built-in validation on mobile and desktop

Interview Question

Q: How do you group radio buttons so only one can be selected?

Ans: Give all the radio inputs in that group the exact same name attribute.

Practice Task

Create a small form using date, range, color, checkbox, and radio inputs.

14. Forms — Labels, Fieldset, Select & Textarea

These tags organise form controls and give users clear, accessible input options.

Tag	Description	Use
<label for=>	Describes an input field	Improves accessibility, tap-to-focus
<fieldset>/<legend>	Groups related controls / caption	Draws a titled box around related fields
<select>/<option>	Dropdown list / one choice	Choosing from fixed options
<textarea>	Multi-line text input	Messages, comments
<datalist>	Suggests options while typing	Autocomplete-style input typing

Course Selection

Choose a course:

Web Development

Message:

Multi-line textarea input...

Email:

name@example.com

<label for="..."> links text ↔ input

Example

```
<fieldset>
  <legend>Course Selection</legend>
  <label for="course">Choose a course:</label>
  <select id="course" name="course">
    <option value="webdev">Web Development</option>
    <option value="ai">AI & Quiz</option>
  </select>
  <label for="msg">Message:</label>
  <textarea id="msg" name="msg" rows="4"></textarea>
</fieldset>
```

Key Points

- label + for improves accessibility and lets users tap the label to focus the field
- fieldset + legend visually and semantically group related fields
- textarea's rows attribute controls its visible height

Interview Question

Q: Why should every input have a linked label?

Ans: It improves accessibility (screen readers announce it) and lets users click the label text to focus the input.

Practice Task

Build a form section using fieldset, legend, a select dropdown, and a textarea.

15. Forms & Validation & Buttons

HTML provides built-in validation attributes and different button types to control form behaviour.

Attribute/Tag	Description	Use
required	Field cannot be left empty	Mandatory fields
minlength/ maxlength, min/ max	Limits text length or numeric range	Username rules, age, quantity
pattern	Custom regex validation	Custom formats like postal codes
disabled / readonly	Locked from editing / visible but not editable	Locking fields, showing pre-filled data
<button type=>	submit / reset / button	Default action, resetting, JS-driven

Registration Form
Username *

Required
Age

min="18" max="60"
Postal Code

pattern="[0-9]{5}" → invalid format

disabled = not focusable, not sent
readonly = visible, still submitted

Example

```
<form>  
  <input type="text" required minlength="3" maxlength="20" placeholder="Username">  
  <input type="number" min="18" max="60" placeholder="Age">  
  <input type="text" pattern="[0-9]{5}" placeholder="Postal Code">  
  <button type="submit">Submit</button>  
  <button type="reset">Reset</button>  
</form>
```

Key Points

- required blocks empty submission automatically, no JavaScript needed
- pattern uses regular expressions for custom rules
- disabled fields are not submitted at all; readonly fields still are

Interview Question

Q: What is the difference between disabled and readonly?

Ans: A disabled field can't be focused or submitted at all, while a readonly field is visible and submitted but can't be edited.

Practice Task

Add required, min/max, and pattern validation to a sample registration form.

16. Semantic HTML Tags

Semantic tags clearly describe their meaning to both the browser and the developer, improving SEO and accessibility.

Tag	Description	Use
<header>	Top section of a page/section	Logo, title, navigation
<nav>	Defines navigation links	Menu, links
<main>	Main content of the document	Only one per page
<section>/<article>	Groups related content / self-contained content	Chapters vs. blog post, news article
<aside>/<footer>	Content aside from main / bottom section	Sidebar, ads / footer info, copyright

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Example

```
<header>
  <h1>My Website</h1>
  <nav><a href="#">Home</a> | <a href="#">About</a></nav>
</header>
<main>
  <section>
    <article><h2>Latest News</h2><p>New course launched!</p></article>
  </section>
  <aside>Quick Links</aside>
</main>
<footer>&copy; 2026 My Website</footer>
```

Key Points

- Semantic tags describe meaning, not just style
- Improve SEO and screen-reader accessibility
- Make code more readable and organized

Interview Question

Q: What is semantic HTML?
Ans: HTML that uses tags which clearly describe the meaning of the content they hold.

Practice Task

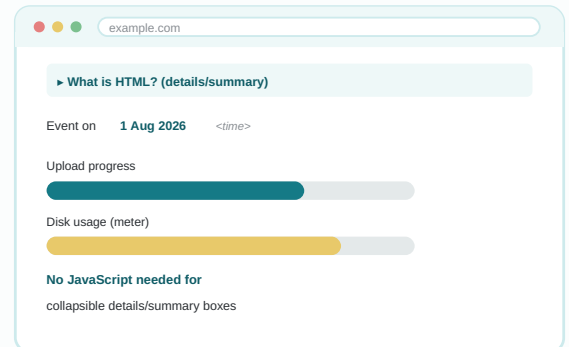
Build a simple webpage layout using all the semantic tags shown above.

17. HTML5 New Semantic & Interactive Elements

HTML5 introduced many new semantic elements that give better structure and add built-in interactivity.

Tag	Description	Use
<details>/<summary>	Collapsible content box / visible label	FAQs, expandable info
<dialog>	Native popup/modal box	Alerts, confirmation dialogs
<time>	Marks up a machine-readable date/time	Event dates, publish dates
<progress>/<meter>	Task completion / measurement in a range	Upload bar / disk usage, ratings
<template>	Holds hidden HTML for later use with JS	Reusable UI chunks

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Example

```
<details>
  <summary>What is HTML?</summary>
  <p>HTML is the standard markup language for web pages.</p>
</details>

<p>Event on <time datetime="2026-08-01">1st August 2026</time></p>
<progress value="70" max="100"></progress>
<meter value="0.8" min="0" max="1"></meter>
```

Key Points

- details/summary need zero JavaScript for a working collapsible box
- time's datetime attribute helps search engines read the real date
- progress = task completion, meter = general measurement/rating

Interview Question

Q: What is the purpose of the details and summary tags?

Ans: They create a native expandable/collapsible content section without needing any JavaScript.

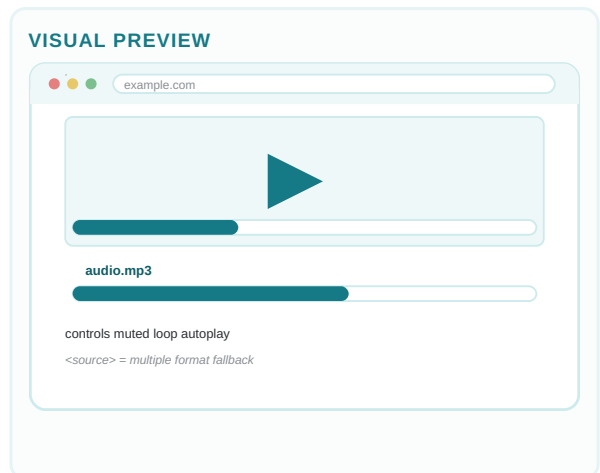
Practice Task

Create an FAQ section using at least 2 details/summary blocks.

18. Media — Audio & Video

HTML lets you embed audio and video directly into a page with native player controls.

Tag	Description	Use
<video controls>	Embeds a video with player controls	Course videos, ads
<audio controls>	Embeds a sound file with player controls	Music, voice notes
<source>	Provides a file source + type	Supports multiple formats/fallback
<track>	Adds subtitles/captions to a video	Accessibility, translations
autoplay / loop / muted	Playback behaviour attributes	muted is required for autoplay in most browsers



Example

```
<video width="320" controls muted>
  <source src="video.mp4" type="video/mp4">
  <track src="subs.vtt" kind="subtitles" srclang="en" label="English">
  Your browser does not support video.
</video>

<audio controls loop>
  <source src="song.mp3" type="audio/mpeg">
</audio>
```

Key Points

- controls attribute shows the play/pause UI
- Always add fallback text for old browsers
- Most browsers block autoplay with sound unless muted is set

Interview Question

Q: What is the purpose of the source tag inside video/audio?

Ans: It lets you provide multiple file formats so the browser picks the first one it supports.

Practice Task

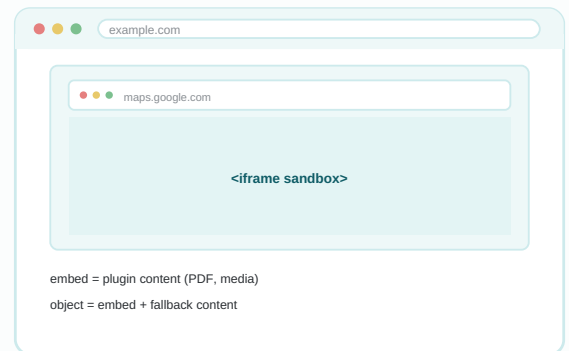
Embed one video and one audio file with controls on a sample page.

19. iframe, embed & object

These tags embed external content — other webpages, PDFs, or media players — directly inside your page.

Tag	Description	Use
<iframe>	Embeds another full webpage	YouTube videos, Google Maps
src / width / height	URL and size of the embedded frame	Points to external content, controls space
<embed>	Embeds external content/ plugins	PDFs, media files
<object>	Embeds external resources with fallback content	PDFs, legacy media
sandbox	Restricts what an iframe can do	Security when embedding untrusted sites

VISUAL PREVIEW



Example

```
<iframe src="https://maps.google.com" width="400" height="250" sandbox></iframe>

<embed src="file.pdf" width="400" height="300" type="application/pdf">

<object data="file.pdf" width="400" height="300">
  PDF cannot be displayed.
</object>
```

Key Points

- iframe is the most common way to embed maps, videos, and forms
- sandbox adds a security layer when embedding third-party content
- object supports fallback content if the resource can't load

Interview Question

Q: Why is the sandbox attribute useful on an iframe?

Ans: It restricts what the embedded page can do (scripts, forms, popups), protecting your site when embedding untrusted content.

Practice Task

Embed a YouTube video and a Google Map on the same practice page using iframe.

20. div, span, class & id

div and span are generic containers used purely for grouping and styling — they carry no semantic meaning of their own.

Item	Description	Use
<div>	Generic block-level container	Grouping content for layout/ CSS
	Generic inline container	Styling a small piece of text
id	Unique identifier	CSS/JS targeting a single element
class	Reusable identifier	Styling multiple elements the same way
#id / .class	CSS selectors targeting id / class	#header {} / .card {}

VISUAL PREVIEW

```
<div class="card" id="profile-card">
  <span>New
  <p>Profile details go here.</p>
</div>

#profile-card { border: 1px solid #ccc; }
```

div = block-level, own line
span = inline, fits in text flow

Example

```
<div class="card" id="profile-card">
  <span class="highlight">New</span>
  <p>Profile details go here.</p>
</div>

<style>
  #profile-card { border: 1px solid #ccc; padding: 10px; }
  .highlight { color: red; font-weight: bold; }
</style>
```

Key Points

- div = block, span = inline
- id is unique per page, class can repeat across many elements
- div/span exist purely for styling and scripting hooks — prefer semantic tags when one fits

Interview Question

Q: What is the difference between div and span?

Ans: div is block-level (own line, full width), span is inline (fits within surrounding text).

Practice Task

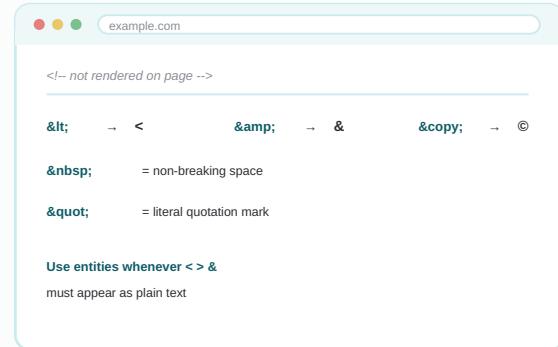
Wrap your portfolio's sections in divs and highlight one word using span, with meaningful id/class names.

21. Comments & HTML Entities

Comments let you leave notes in code without displaying them, and entities let you safely display reserved or special characters.

Item	Description	Example
<code><!-- comment --></code>	Not rendered on the page	Notes for developers, disabling code
<code>&lt; / &gt;</code>	Displays a literal <code>< / ></code> symbol	Showing HTML code as text
<code>&amp;</code>	Displays a literal <code>&</code> symbol	Ampersands in text
<code>&nbsp;</code>	Non-breaking space	Prevents line break at that space
<code>&copy; / &quot;</code>	Copyright symbol / literal quotation mark	Footer credits, quoting attributes

VISUAL PREVIEW



Example

```
<!-- This is the main navigation, do not remove -->
<nav>...</nav>

<p>Use &lt;p&gt; to create a paragraph.</p>
<p>&copy; 2026 Inquisitors&nbsp;Society</p>
```

Key Points

- Comments are visible in the page source but never rendered visually
- Entities are required whenever you need to show `< > &` as literal text
- ` ` keeps two words glued together on one line

Interview Question

Q: Why can't you just type `<` directly into your HTML text?

Ans: The browser would interpret it as the start of a tag; entities like `<` are needed to display it as plain text.

Practice Task

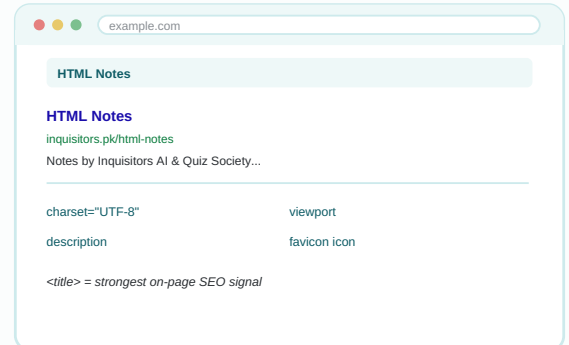
Write a short paragraph that displays the text `<body>` using the correct HTML entities.

22. Meta Tags & SEO Basics

Meta tags live in the head section and give browsers and search engines extra information about the page.

Tag	Description	Use
<meta charset>	Sets character encoding	Always UTF-8 for modern sites
<meta name="viewport">	Controls mobile scaling	Required for responsive design
<meta name="description">	Short page summary	Shown in search engine results
<meta name="author">	Names the page's author	Credit/attribution
<link rel="icon" /> / <title>	Favicon / page title shown in tab	Brand identity / strongest on-page SEO tag

VISUAL PREVIEW



Example

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Notes by Inquisitors AI & Quiz Society">
  <meta name="author" content="Inquisitors Society">
  <link rel="icon" href="favicon.ico">
  <title>HTML Notes</title>
</head>
```

Key Points

- The description meta tag directly affects your search engine snippet
- viewport meta tag is mandatory for any responsive/mobile-friendly site
- A clear, unique title tag is one of the strongest on-page SEO signals

Interview Question

Q: Why is the meta description tag important for SEO?

Ans: It's often shown as the summary text under your page's link in search engine results, influencing click-through rate.

Practice Task

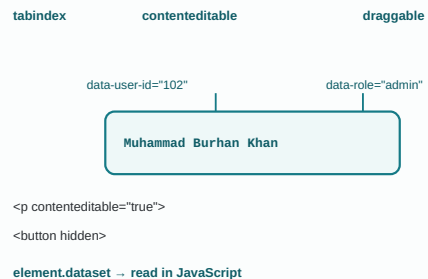
Write a complete head section for a portfolio page including charset, viewport, description, and favicon.

23. Global & data-* Attributes

Beyond id, class, and style, HTML supports several other global attributes and custom data attributes for storing extra info.

Attribute	Description	Use
tabindex	Controls keyboard tab order	Accessibility, custom focus order
contenteditable	Makes an element editable in the browser	Live text editors
draggable	Makes an element drag-and-drop capable	Interactive UI, sortable lists
hidden	Hides an element completely	Toggle visibility with JS
data-*	Stores custom data on an element	Passing data to JavaScript

VISUAL PREVIEW



Example

```
<div data-user-id="102" data-role="admin">
  Muhammad Burhan Khan
</div>

<p contenteditable="true">Click here and start typing...</p>

<button hidden>You can't see me yet</button>
```

Key Points

- data-* attributes are custom and can be named anything after 'data-'
- JavaScript reads data-* attributes via element.dataset
- hidden instantly removes an element from the visible page

Interview Question

Q: What is the purpose of data-* attributes?

Ans: They let developers store extra custom information directly on an HTML element, to be read and used later by JavaScript.

Practice Task

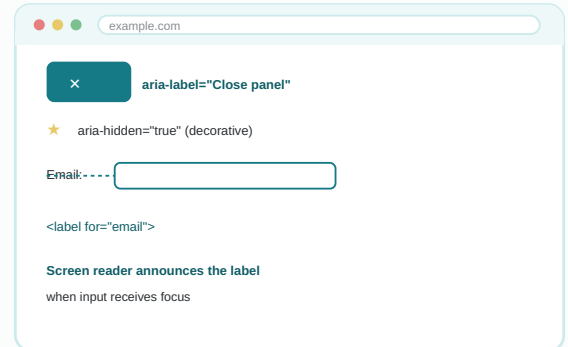
Add two custom data-* attributes to a div and explain what each one could be used for.

24. Accessibility (ARIA) in HTML

Accessible HTML makes sure every user, including those using screen readers, can understand and use your page.

Attribute/Tag	Description	Use
alt	Text alternative for images	Screen readers, broken image fallback
aria-label	Accessible name for an element	Icons/buttons without visible text
aria-hidden="true"	Hides decorative content from screen readers	Icons that add no real information
role	Defines what an element represents	role="button", role="navigation"
Semantic tags / label+for	Built-in accessible structure / connects text to a field	header/nav/main/footer / screen reader announces label

VISUAL PREVIEW



Example

```
<button aria-label="Close panel">&times;</button>
<span aria-hidden="true">&#9733;</span>
<nav role="navigation">...</nav>

<label for="email">Email:</label>
<input id="email" type="email">
```

Key Points

- Always prefer real semantic tags before reaching for ARIA roles
- aria-label is essential for icon-only buttons
- Accessible HTML benefits everyone, not only users of assistive tech

Interview Question

Q: Why is aria-label needed on an icon-only button?

Ans: Because a screen reader can't read an icon; aria-label gives it text to announce instead.

Practice Task

Add aria-label to two icon buttons on a sample page (e.g. a search icon and a close icon).

25. Introduction to Canvas & SVG

HTML also supports drawing graphics directly in the browser using the canvas and SVG elements.

Tag	Description	Use
<canvas>	A blank drawing surface controlled by JavaScript	Games, charts, dynamic graphics
<svg>	Scalable Vector Graphics, written directly in HTML	Icons, logos, resolution-independent graphics
SVG shapes	circle, rect, line, path, etc.	Building vector illustrations
Canvas vs SVG	Canvas = pixels (JS-drawn); SVG = shapes (DOM)	Choose based on complexity & interactivity needed

VISUAL PREVIEW

<canvas>



pixels, JS-drawn

<svg>



vector, DOM shapes

SVG scales perfectly at any size;
canvas can blur when resized

Example

```
<canvas id="myCanvas" width="200" height="100"></canvas>
<script>
  const ctx = document.getElementById("myCanvas").getContext("2d");
  ctx.fillStyle = "teal";
  ctx.fillRect(10, 10, 100, 50);
</script>

<svg width="100" height="100">
  <circle cx="50" cy="50" r="40" fill="orange"></circle>
</svg>
```

Key Points

- canvas needs JavaScript to draw anything; it's just a blank box in pure HTML
- SVG shapes are real elements in the page, so they can be styled with CSS
- SVG scales perfectly at any size; canvas can get blurry when resized

Interview Question

Q: What is the main difference between canvas and SVG?

Ans: Canvas is pixel-based and drawn with JavaScript, while SVG is made of vector shapes that live as elements in the DOM.

Practice Task

Draw one simple shape using SVG (e.g. a circle or rectangle) with a fill color of your choice.

26. HTML Best Practices & Common Mistakes

Writing clean, valid HTML makes your pages easier to maintain, more accessible, and better for SEO.

Practice	Why it Matters
Always close your tags	Unclosed tags can break page structure in some browsers
Use lowercase tag & attribute names	Keeps code consistent and readable
Indent nested elements	Makes the code structure easy to follow
Never skip alt on images	Required for accessibility and SEO
Use semantic tags over div soup	Improves SEO, accessibility, and readability
One h1 per page & validate your HTML	Clear content outline; catches typos and errors early

VISUAL PREVIEW

✓ Good

```
<article><h2>Title</h2>
<img alt="cover image">
properly indented, semantic
```

✗ Avoid

```
<div><div><img><div>Title
unclosed tags, missing alt, div soup
```

Always close tags • lowercase names

one <h1> per page • validate HTML

Example

```
<!-- Good practice -->
<article>
  <h2>Blog Post Title</h2>
  <p>Well-structured, indented, semantic HTML.</p>
  
</article>

<!-- Avoid: unclosed tags, missing alt, everything as div -->
<div><div><div>Title
</div>
```

Key Points

- Clean HTML is easier to style, debug, and maintain long-term
- Semantic + accessible HTML directly improves SEO ranking
- Small habits (closing tags, alt text, indentation) prevent big bugs later

Interview Question

Q: Why is 'div soup' considered bad practice?

Ans: Overusing divs for everything removes semantic meaning, hurting accessibility and SEO compared to proper semantic tags.

Practice Task

Review one of your earlier practice pages and fix at least 3 things using these best practices.



Thank You!

These notes cover everything needed for the Week 1 HTML task, from the basics to advanced concepts, now paired with a visual diagram on every page. Keep practicing, keep building — and reach out anytime you're stuck.

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